

Class 14: Information Theory

- **Part 1: Continuation of Shannon's paper**
- **Part 2: Determining authorship of *The Federalist* by Mosteller and Wallace**

Continuing Shannon's paper

Shannon's key idea

- **A discrete information source is characterized by the probabilities for emitting signals from the source**

To give a visual idea of how this series of processes approaches a language, typical sequences in the approximations to English have been constructed and are given below. In all cases we have assumed a 27-symbol “alphabet,” the 26 letters and a space.

1. Zero-order approximation (symbols independent and equiprobable).

XFOML RXKHRJFFJUJ ZLPWCFWKCYJ FFJEYVKCQSGHYD QPAAMKBZAACIBZL-
HJQD.

2. First-order approximation (symbols independent but with frequencies of English text).

OCRO HLI RGWR NMIELWIS EU LL NBNESEBYA TH EEI ALHENHTTPA OOBTTVA
NAH BRL.

3. Second-order approximation (digram structure as in English).

ON IE ANTSOUTINYS ARE T INCTORE ST BE S DEAMY ACHIN D ILONASIVE TU-
COOWE AT TEASONARE FUSO TIZIN ANDY TOBE SEACE CTISBE.

4. Third-order approximation (trigram structure as in English).

IN NO IST LAT WHEY CRATICT FROURE BIRS GROCID PONDENOME OF DEMONS-
TURES OF THE REPTAGIN IS REGOACTIONA OF CRE.

5. First-order word approximation. Rather than continue with tetragram, $\dots$, n -gram structure it is easier and better to jump at this point to word units. Here words are chosen independently but with their appropriate frequencies.

REPRESENTING AND SPEEDILY IS AN GOOD APT OR COME CAN DIFFERENT NAT-
URAL HERE HE THE A IN CAME THE TO OF TO EXPERT GRAY COME TO FURNISHES
THE LINE MESSAGE HAD BE THESE.

6. Second-order word approximation. The word transition probabilities are correct but no further structure is included.

THE HEAD AND IN FRONTAL ATTACK ON AN ENGLISH WRITER THAT THE CHAR-
ACTER OF THIS POINT IS THEREFORE ANOTHER METHOD FOR THE LETTERS THAT
THE TIME OF WHO EVER TOLD THE PROBLEM FOR AN UNEXPECTED.

How to simulate the digram structure

3. Second-order approximation (digram structure as in English).

ON IE ANTSOUTINYS ARE T INCTORE ST BE S DEAMY ACHIN D ILONASIVE TU-
COOWE AT TEASONARE FUSO TIZIN ANDY TOBE SEACE CTISBE.

To construct (3) for example, one opens a book at random and selects a letter at random on the page. This letter is recorded. The book is then opened to another page and one reads until this letter is encountered. The succeeding letter is then recorded. Turning to another page this second letter is searched for and the succeeding letter recorded, etc. A similar process was used for (4), (5) and (6). It would be interesting if further approximations could be constructed, but the labor involved becomes enormous at the next stage.

Measuring information content

Suppose we have syllables with probabilities $p_1, p_2, \dots, p_N$

How much choice do we have in the selection a word?

How uncertain are we of the outcome?

Entropy – measure of disorder or uncertainty in a system

Maximum entropy example

toss a fair coin many times; no way to predict outcome of next toss

Minimum entropy example

toss a two-tailed coin many times; outcome is certain, next toss is tails

Measuring information content

Shannon Entropy

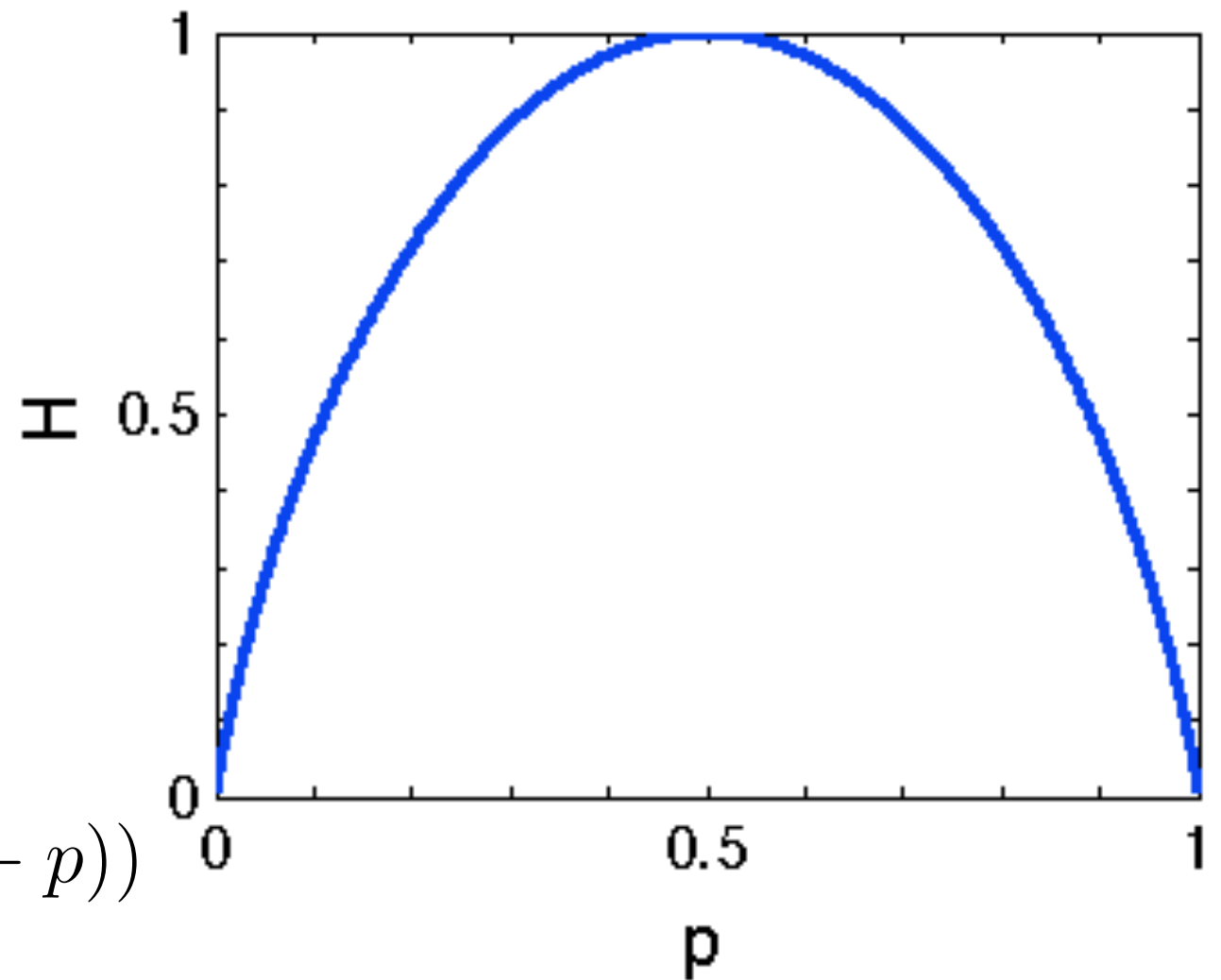
$$H = - \sum_{i=1}^N p_i \log(p_i)$$

Example ($N = 2$)

$$p_1 = p$$

$$p_2 = 1 - p_1 = 1 - p$$

$$H = - (p \log p + (1 - p) \log(1 - p))$$



$H = 0$ only if $p = 0$ or $p = 1$. Does this make sense?

H attains its maximum value at $p = 1/2$. Does this make sense?

Measuring information content

Shannon Entropy

$$H = - \sum_{i=1}^N p_i \log(p_i)$$

Measures the average information content of a source

When you utter a word, how many other choices of words are you choosing between? Can you calculate this?

Shannon's fundamental theorem

9. THE FUNDAMENTAL THEOREM FOR A NOISELESS CHANNEL

We will now justify our interpretation of H as the rate of generating information by proving that H determines the channel capacity required with most efficient coding.

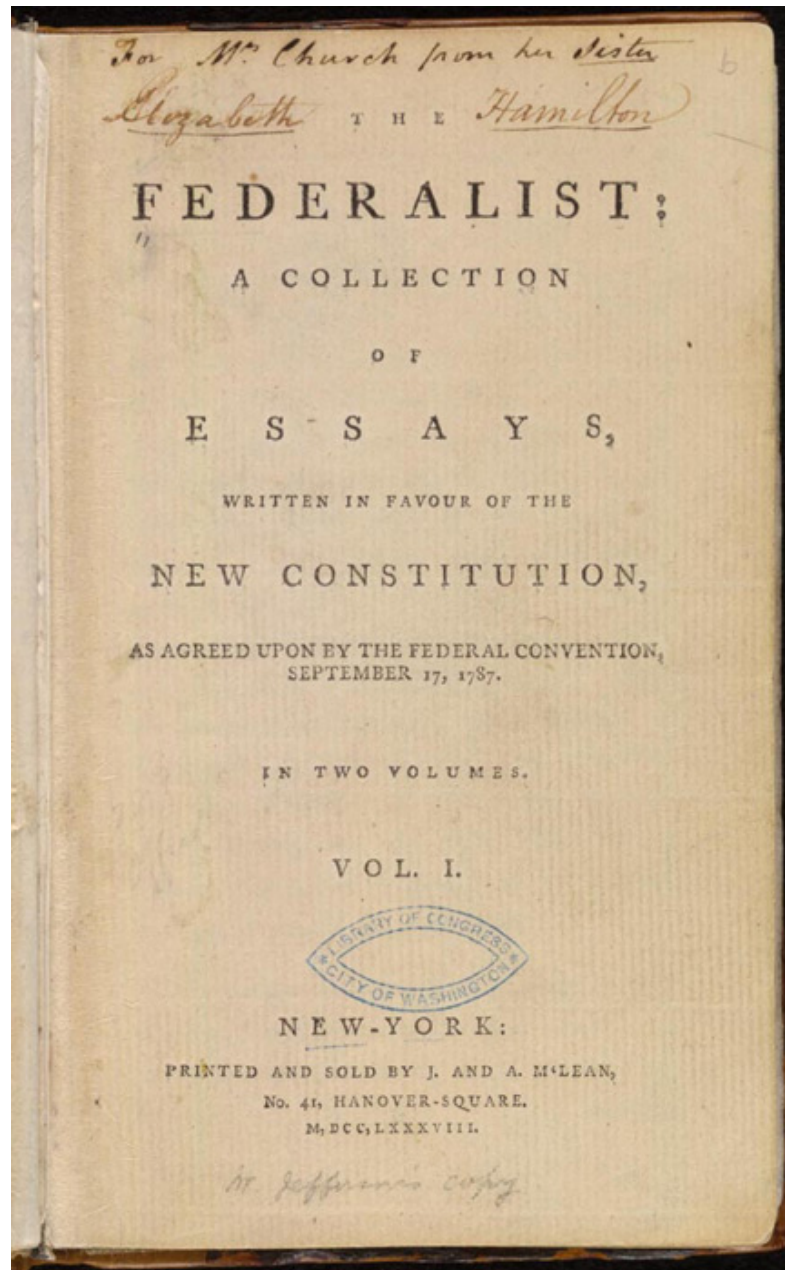
Theorem 9: Let a source have entropy H (bits per symbol) and a channel have a capacity C (bits per second). Then it is possible to encode the output of the source in such a way as to transmit at the average rate $\frac{C}{H} - \epsilon$ symbols per second over the channel where ϵ is arbitrarily small. It is not possible to transmit at an average rate greater than $\frac{C}{H}$.

- **Suppose the information source has entropy H and the channel has capacity C**
- **One can encode the output of the source so that it can be transmitted a rate of up to C/H symbols per second**
- **It is not possible to transmit faster than C/H !!**

Historical mystery

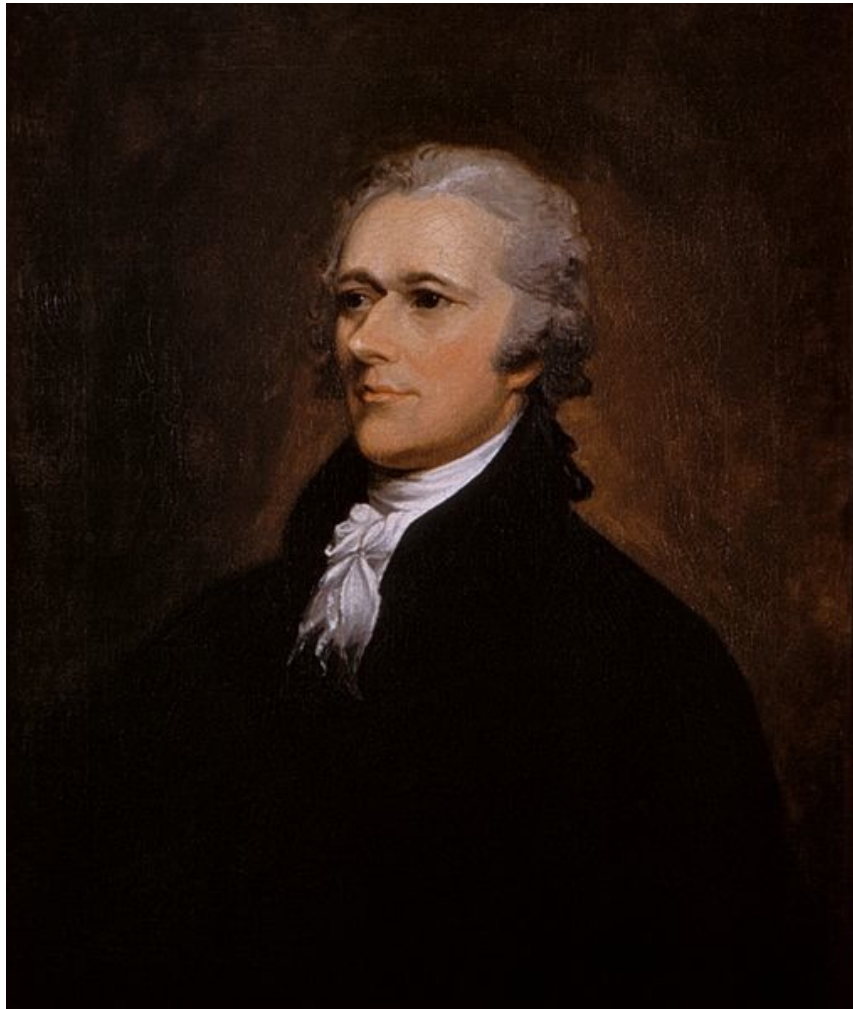
Who wrote the Federalist papers?

The Federalist

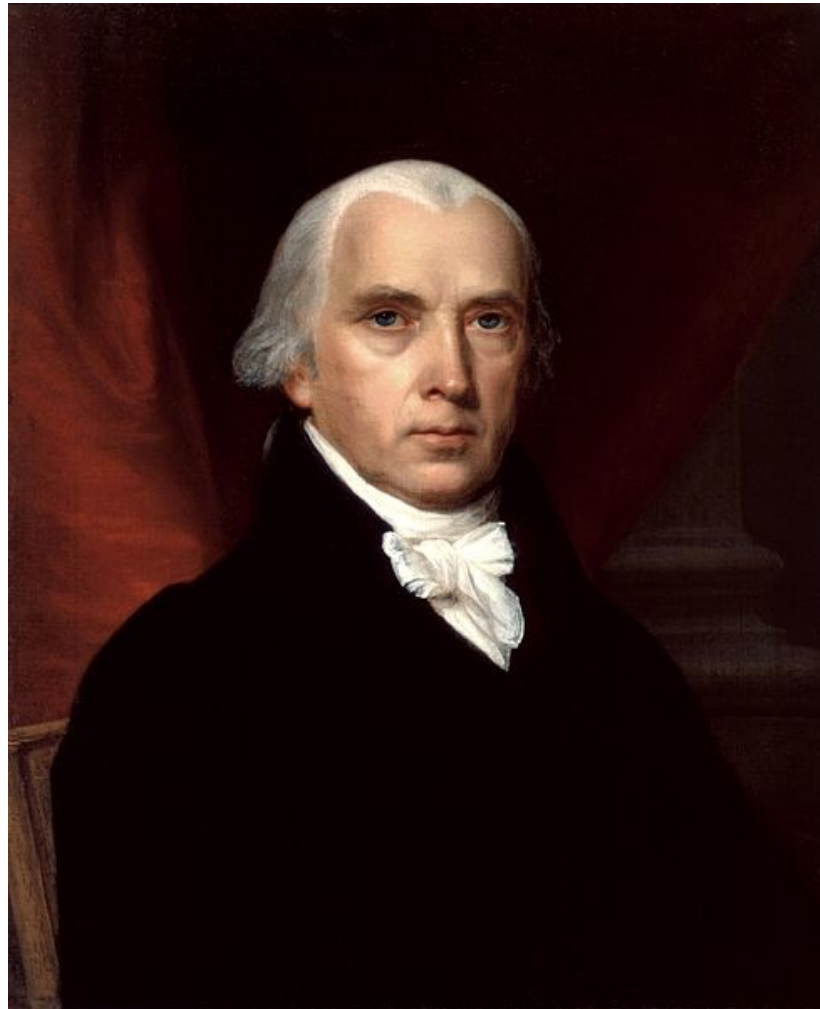


- 85 articles published in New York advocating the ratification of the US constitution during 1787–1788
- Used in many court cases as a way to interpret the Founding Fathers' intentions about the constitution
- Authorship was a closely guarded secret (written under the pseudonym *Publius*)

Suspected authors



Alexander Hamilton
(1757–1804)



James Madison
(1751–1836)



John Jay
(1745–1829)

Disputed authorship

- Soon before his [fatal gun duel](#), Hamilton provided a list claiming authorship of 63 of the articles (with three jointly written with Madison)
- Later, Madison claimed 29 for himself, suggesting that the difference was “owing doubtless to the hurry in which [Hamilton’s] memorandum was made out”
- Hamilton’s list contained at least one known error, attributing No. 54 to Jay, when in fact Jay wrote No. 64
- How can the authorship of the articles be determined?

An information theory approach

by Frederick Mosteller and David L. Wallace

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INFERENCE IN AN AUTHORSHIP PROBLEM^{1,2}

A comparative study of discrimination methods applied
to the authorship of the disputed *Federalist* papers

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Inference and Disputed Authorship:

The Federalist

1.1. PURPOSE

When two statisticians, both flanks unguarded, blunder into an historical and literary controversy, merciless slaughter is imminent. Our persistence needs explanation.

From the point of view of statistical methods, authorship problems fall into a general area called **discrimination or classification problems**. In these problems the task is to assign a category to an object or individual whose true category is uncertain. In our authorship problem the objects are essays written by either Hamilton or Madison. We reduce our uncertainty about the authorship of an “unknown” essay by comparing its properties with information obtained from essays whose authorship is known. Classifying plants in biology, skulls in anthropology, candidates for parole in criminology, and subjects according to personality in psychology are related operations that sometimes employ similar methods, even though the properties that aid the classification vary drastically from one area to another. The methods used to study one problem in discrimination can sometimes be extended to other areas of research. We are concerned with the methodology of discrimination studies, and we especially wish to compare a number of methods of discrimination all based on much the same data.

Approach #1: Wordiness

Williams and Mosteller independently counted the number of words in all the sentences of *The Federalist*,* and their first frustration was the discovery of an important empirical principle—people cannot count, at least not very high (Miller, 1956; Wundt, 1912). But they created checks to get accurate results. The second difficulty, common to nearly every investigation, is that special problems must be settled by judgment. How should quotations from other authors be handled? How should numbers written out or in numerals be counted in words, and so on? These matters took a great deal of time, but finally, the task was done. When Williams and Mosteller emerged from their bout with the desk calculators, their second frustration was:

Average sentence length: Hamilton—34.55 (words),
Madison—34.59 (words).

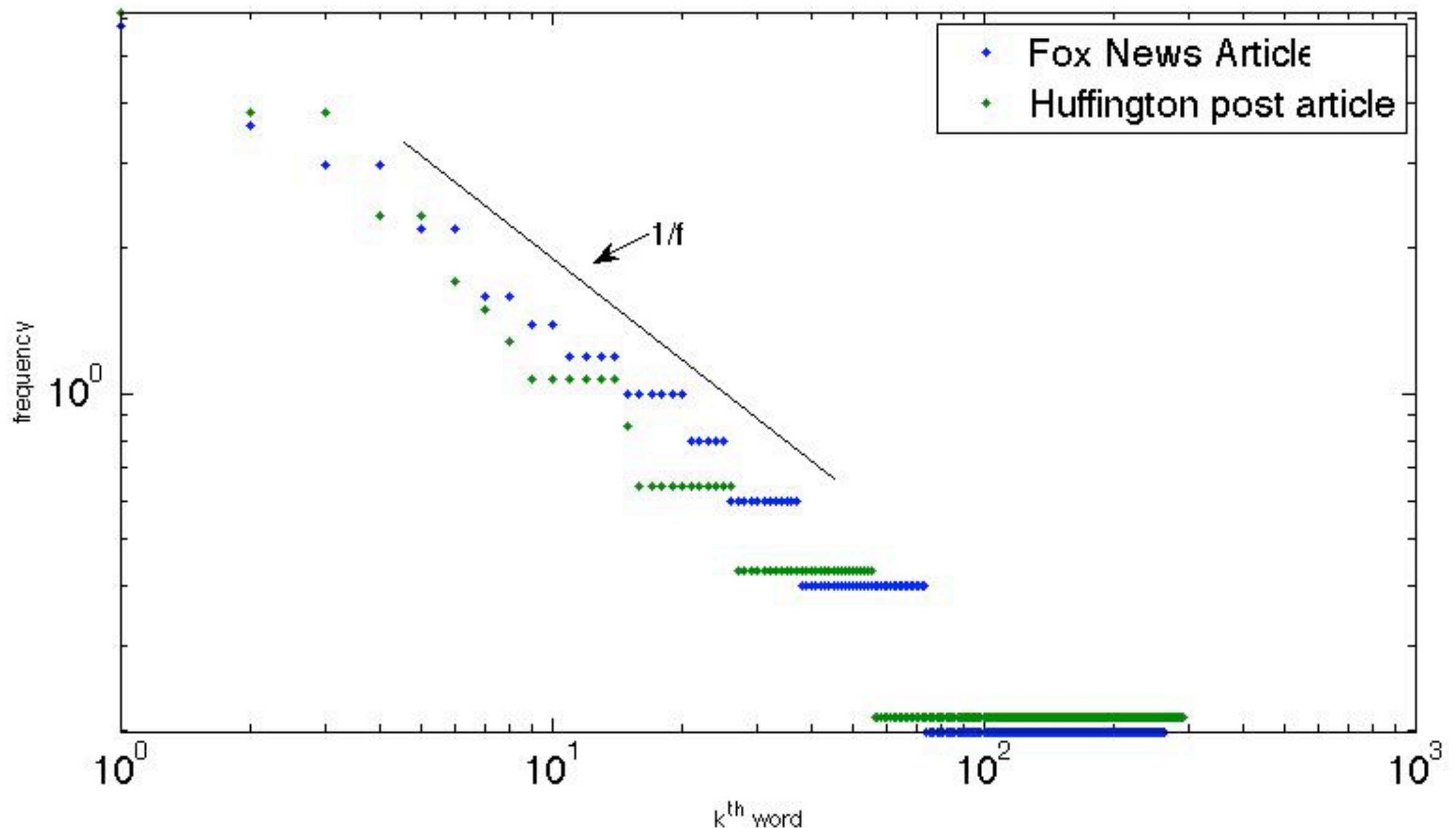
Still all was not lost: perhaps the variability in sentence length was the key to the style. They computed the standard deviation (a measure of variability) of the sentence lengths for each paper, then found the average of these for the two men:

Average standard deviation: Hamilton—19.2 (words),
Madison—20.3 (words).

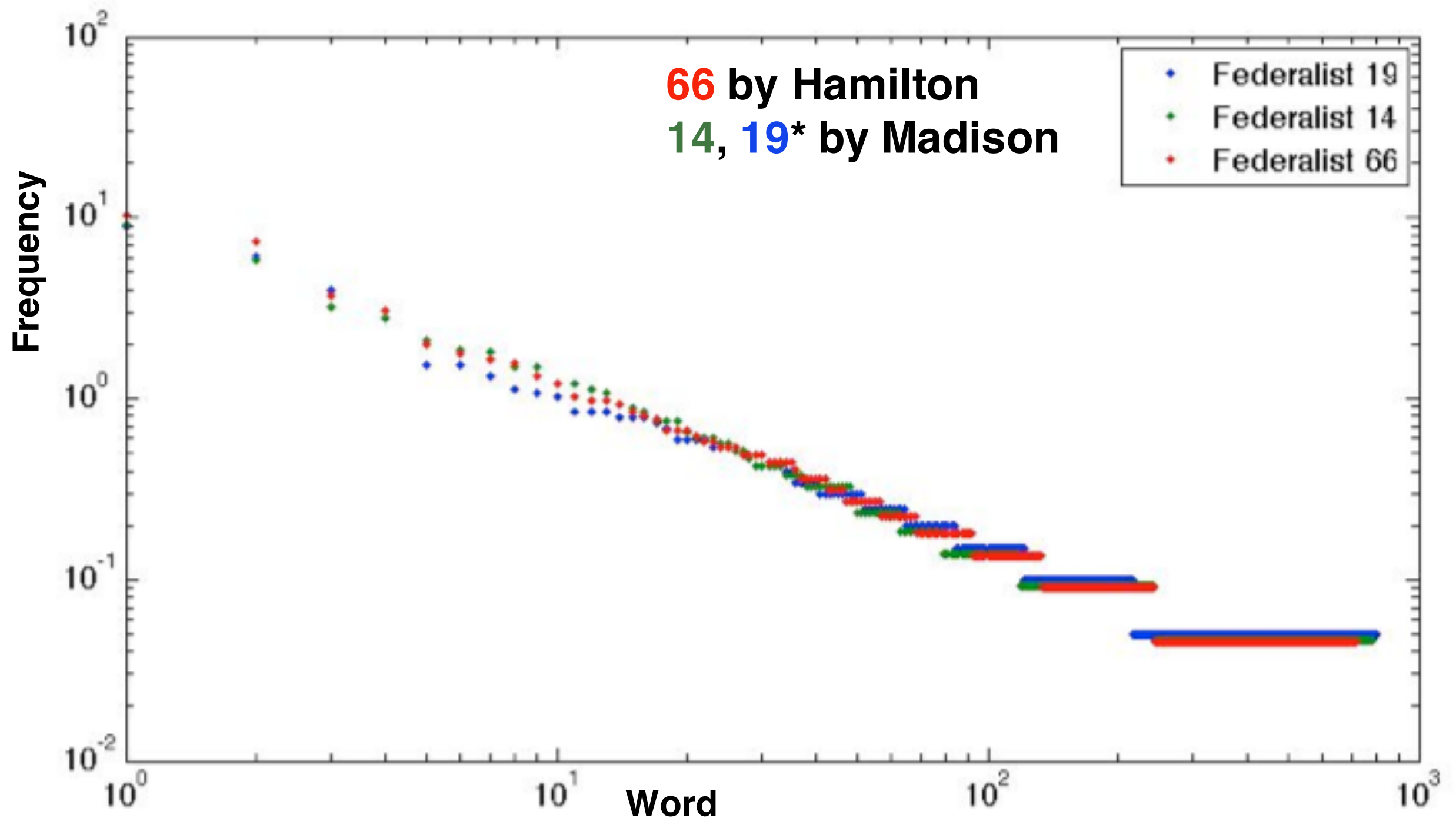
Needless to say, these measures would be hopeless for discriminating between the two authors.

Approach #2: Zipf's law

Frequency of word is inversely proportional to rank in frequency table



Approach #2: Zipf's law



Approach #2: Zipf's law

Cooke (1961, p.xxviii) says of the remarkably similar prose styles of Hamilton and Madison:*

“This was no unique phenomenon, for most educated Americans of the late eighteenth century—a few particularly gifted writers, like Jefferson, perhaps excepted—employed the same stylistic devices, the same standard phrases, and remarkably similar sentence structure.”

**Applied Bayesian and Classical Inference: The Case of
The Federalist Papers**
By F. Mosteller, D. L. Wallace

Approach #3: look for special words

About 1959, after previous correspondence, Douglass Adair informed Mosteller that he had found a pair of words (which we call marker words) that distinguished Hamilton and Madison quite well when the words occurred. Hamilton uses *while* and Madison in a corresponding situation uses *whilst*. Adair has pursued his investigation of these words, including verification in the original newspapers. He has also found an earlier writer, Bailey (1916), who noticed the *while-whilst* distinction. On the other hand, both words have low rates per thousand words of text, so that not all papers can be discriminated by them.

- **Experts in the field had considered this, but did not see how to use this to determine authorship with certainty**
- **But Mosteller and Wallace pursued it in the hope that with enough markers the evidence would become overwhelming**

Mosteller and Wallace's approach

Adair's findings encouraged us to look for additional marker words and to begin a large study. We knew we wanted the counts done on high-speed computers, but while the programmers worked, we decided to have word counts made on a few papers by hand, as described in Section 2.6.

We then made a rough screening plan that involved looking at incidences of each word in 6 Hamilton and 5 Madison papers (we call this set wave A); this screening led to the elimination or choice of marker word candidates. Straightforward objective rules were used, except that contextual words (see Section 2.1) were discarded freely. We then tried the words that looked promising on a set of papers on similar topics (5 by Hamilton and 5 by Madison), labeled the *Exterior set*, not in *The Federalist* (there is danger in the screening process of wasting the known Madison *Federalist* papers). This led to further elimination of marker word candidates. By now, Adair's *while-whilst* pair had emerged as important candidates (as we already knew), thereby suggesting that the screening method was not hopeless. Another word, *upon*, emerged as a Hamilton marker. These three words were tried on a second set of 10 *Federalist* papers (6 Hamilton and 4 Madison), wave B, and *upon* by itself separated the 10 papers correctly and emphatically because Hamilton always had a rate of use higher than a prechosen number, about 1.5 per 1000 words, and Madison always a lower rate, usually zero. We decided that *upon* was definitely a marker word.

Tabulated data

TABLE 1.4-1

INCIDENCE: NUMBERS OF PAPERS IN WHICH WORD OCCURRED
AT LEAST ONCE

	<i>enough</i>	<i>while</i>	<i>whilst</i>	<i>upon</i>	Number of papers examined
Hamilton	14	10	0	23	23
Madison	0	0	13	4	19
Disputed	0	0	5	1	12
Joint	1	0	2	2	3

TABLE 1.4-2

RATES PER 1000 WORDS

	<i>enough</i>	<i>while</i>	<i>whilst</i>	<i>upon</i>	Total words in 1000's
Hamilton	0.59	0.26	0	2.93	45.7
Madison	0	0	0.47	0.16	51.0
Disputed	0	0	0.34	0.08	23.9
Joint	0.18	0	0.36	0.36	5.5
					<u>126.1</u>

Systematic study

Hamilton

TABLE 1.4-3
HAMILTON KNOWN PAPERS

Paper	Wave	Total words	enough	while	whilst	upon	Rate per 1000 words upon
<i>Federalist</i>							
1	A	1,570	1	—	—	6	3.82
6	A	1,893	1	—	—	4	2.11
7	A	2,245	—	—	—	11	4.90
8	A	1,986	2	—	—	3	1.51
9	A	1,629	1	1	—	4	2.46
11	B	2,495	—	—	—	6	2.40
12	B	2,140	1	—	—	7	3.27
24	B	1,805	—	1	—	7	3.88
26	C	2,379	7	—	—	6	2.52
36	B	2,721	—	—	—	6	2.21
59	C	1,847	—	—	—	3	1.62
65	B	2,013	1	1	—	10	4.97
66	A	2,215	1	1	—	11	4.97
70	B	3,029	3	2	—	6	1.98
71	C	1,702	3	1	—	3	1.76
72	C	2,032	2	2	—	5	2.46
74	C	906	—	—	—	2	2.21
77	C	1,970	1	1	—	10	5.08
		36,577					
Exterior (identified in Section 2.2)							
Cont. 1		1,188	—	—	—	2	1.68
Cont. 2		1,380	2	—	—	5	3.62
Pac. 1		2,945	—	1	—	6	2.04
Pac. 2		2,470	1	1	—	10	4.05
Pac. 3		1,178	—	—	—	1	.85
		9,161					
Total		45,738	27	12	0	134	
Rate per 1000 words:			.59	.26	0	2.93	

Madison

TABLE 1.4-4
MADISON KNOWN PAPERS

Paper	Wave	Total words	enough	while	whilst	upon	Rate per 1000 words upon
<i>Federalist</i>							
10	A	2,987	—	—	—	—	—
14	A	2,149	—	—	1	—	—
37	A	2,709	—	—	1	1	.37
38	A	3,314	—	—	2	4	1.21
39	C	2,585	—	—	—	—	—
40	B	2,718	—	—	—	—	—
41	B	3,506	—	—	1	—	—
42	C	2,711	—	—	—	2	.74
43	C	3,029	—	—	1	—	—
44	B	2,561	—	—	2	—	—
45	C	2,117	—	—	2	—	—
46	B	2,601	—	—	2	—	—
47	A	2,547	—	—	—	—	—
48	C	1,561	—	—	—	—	—
		37,095					
Exterior (identified in Section 2.2)							
Helv. 1		2,945	—	—	1	—	—
Helv. 2		2,704	—	—	3	1	.37
Helv. 3		2,458	—	—	1	—	—
N. Am. 1		3,048	—	—	3	—	—
N. Am. 2		2,798	—	—	4	—	—
		13,953					
Total		51,048	—	—	24	8	
Rate per 1000 words:			0	0	.47	.16	

TABLE 1.4-5
JOINT AND DISPUTED PAPERS

Paper	Total words	Absolute counts				Rate per 1000 words
		<i>enough</i>	<i>while</i>	<i>whilst</i>	<i>upon</i>	<i>upon</i>
Joint						
18	2,084	1	—	1	1	.48
19	2,019	—	—	1	—	—
20	1,438	—	—	—	1	.70
Total	5,541	1	—	2	2	
Rate per 1000 words:		.18	0	.36	.36	
Disputed						
49	1,594	—	—	1	—	—
50	1,103	—	—	—	—	—
51	1,911	—	—	2	—	—
52	1,841	—	—	—	—	—
53	2,160	—	—	1	—	—
54	1,996	—	—	—	2	1.00
55	2,034	—	—	—	—	—
56	1,560	—	—	1	—	—
57	2,200	—	—	3	—	—
58	2,082	—	—	—	—	—
62	2,380	—	—	—	—	—
63	3,020	—	—	—	—	—
Total	23,881	0	0	8	2	
Rate per 1000 words:		0	0	.34	.08	

On the basis of Table 1.4-5 the evidence seems to be that all 12 disputed papers are by Madison and that most of the material in Nos. 18-20 is too. For us the problem is to extend this evidence and to measure its strength. We are not faced with a black-or-white situation, and we are not going to provide an absolutely conclusive settlement. Outside those parts of logic and mathematics that are divorced from the empirical world, strong confidence in conclusions is the most an investigation can be expected to offer.

Getting more specific

- **How can we ascribe an actual probability that a given paper is written by Hamilton and Madison?**
- **A random variable X follows a Poisson distribution if**

$$P(X = n) = \frac{e^{-\mu} \mu^n}{n!}$$

- **A good model? The total number of occurrences of a particular word in a given paper follows a Poisson distribution**

Bayes' Theorem

- Let A be the event that you have a rare disease, and B be the probability that you test positive for this disease
- The probability of

$$P(A, B) = P(A)P(A|B)$$

Joint probability of
both events

Probability of A
happening given that
 B happened

- Similarly, $P(B, A) = P(B)P(B|A)$ and hence

$$P(A|B) = \frac{P(B|A)}{P(B)} P(A)$$

Posterior – my estimate of
having disease, given the
positive test result

Prior – my initial
estimate of
having disease

Example: “also”

TABLE 3.1-1

POISSON PROBABILITIES WITH EXPECTED COUNTS

$$w\mu_H = .62, w\mu_M = 1.34$$

Expected number of occurrences

w = word count in 1000s

μ = rate per 1000 words

2000 total words

Frequency	Hamilton	Madison
0	.538	.262
1	.334	.351
2	.103	.235
3	.0214	.105
4	.00331	.0352
5	.000411	.00943
6	.0000424	.00211

- Let H be the event that Hamilton wrote the paper, and let M be the event that Madison wrote the paper
- Suppose that “also” is used 4 times a paper. What are the odds that Hamilton wrote the paper?

$$P(X = 4|H) = \frac{e^{-\mu_H} \mu_H^4}{4!} = 0.00331 \quad \mu_H = 0.31$$

$$P(X = 4|M) = \frac{e^{-\mu_M} \mu_M^4}{4!} = 0.0352 \quad \mu_M = 0.67$$

Two interpretations

- **Ratio of probabilities (“likelihood ratio”)**

$$\frac{P(X = 4|H)}{P(X = 4|M)} \approx \frac{1}{10}$$

- **Hence you might say that the odds are 10:1 that Madison wrote the paper.**
- **But this depends on what your prior belief about who wrote the papers is.**
- **If you had other reasons to think Hamilton wrote the paper, you might feel differently.**

The use of Bayes' Theorem

- **From Bayes' Theorem**

$$P(H|X) = \frac{P(X|H)}{P(X)} P(H)$$

$$P(M|X) = \frac{P(X|M)}{P(X)} P(M)$$

- **Hence**

$$\frac{P(H|X)}{P(M|X)} = \frac{P(X|H)}{P(X|M)} \frac{P(H)}{P(M)}$$

Posterior odds of
Hamilton versus
Madison

Probabilities
estimated from
Poisson model and
observed data

Prior odds of
Hamilton versus
Madison

3.1B. Selecting words and weighting their evidence. Bayes' theorem, as described in Eq. (1), provides one rationale for selecting the words that we use to determine authorship, and it provides automatically for weighting the evidence from each word, as we now illustrate.

TABLE 3.1-2

RATES PER THOUSAND WORDS FOR *also*, *an*, AND *because*,
AND A MEASURE OF IMPORTANCE FOR DISCRIMINATION

Word	Hamilton rate	Madison rate	Importance ($w = 2$)
<i>also</i>	.31	.67	.55
<i>an</i>	6.00	4.50	.86
<i>because</i>	.45	.50	.01

Let us extend our previous example and suppose that the counts for *also*, *an*, and *because* are independent Poisson variables, with the known rates given in Table 3.1-2. Suppose that a 2000-word paper by one of our two authors contains 4 *also*'s, 7 *an*'s, and 0 *because*'s. If Hamilton is the author, the probability of the triple (4, 7, 0) is the product of three Poisson probabilities

$$f_P(4 | .62)f_P(7 | 12)f_P(0 | .9),$$

and if Madison is the author, the probability is

$$f_P(4 | 1.34)f_P(7 | 9)f_P(0 | 1).$$

Finally, the likelihood ratio for the three words is the product of the likelihood ratios for the separate words, and the weighting problem is solved.

The numerical values of the Poisson probabilities instruct us further. They give the likelihood ratio as

$$\frac{.00331}{.0352} \times \frac{.0437}{.117} \times \frac{.407}{.368} \approx .0940 \times .374 \times 1.11 \approx .039.$$

$$\frac{P(X|H)}{P(X|M)}$$

$$P(X|M)$$

$$\frac{P(X|H)}{P(X|M)}$$

Subtleties

$$p_1 = P(H) \quad p_2 = P(M)$$

3.1C. Initial odds. All this seems perfectly straightforward. Wherein lie the difficulties? The first place is in the choice of the *initial odds*—here, the values of p_1 and p_2 . Your final odds—posterior odds—will differ from mine, if we choose differing values for p_1 . In some problems, the choice of p_1 might be made on the basis of objective frequencies, but in others, personal degree of belief may be involved. In our own problem, you might regard the initial odds as 1 for an unknown paper, thus setting a 50-50 chance as appropriate to your degree of ignorance. Or you might notice that Hamilton wrote 43 of the known papers to Madison's 14, and then assign prior odds of $43/14 \approx 3$. Or if you are an historian with knowledge of the problem, you may have quite strong beliefs that lead to the assignment of large odds in one or the other direction.

$$\frac{P(H|X)}{P(M|X)} = \frac{P(X|H)}{P(X|M)} \frac{P(H)}{P(M)}$$

Conclusions

(1) Our data independently supplement that of the historian. On the basis of our data alone, Madison is extremely likely, in the sense of degree of belief, to have written the disputed *Federalists*, with the possible exception of No. 55, and there our evidence is weak, suitable deflated odds are 80 to 1 for Madison. No. 56, next weakest, is a strong 800 to 1 for Madison. The data are strong for all the rest, including the two papers historians feel weakest about, Nos. 62 and 63.

(2) Among the joint papers, Nos. 18 and 19 look as if Madison wrote the lion's share. No. 20 requires a more subtle analysis of its possible contamination by third parties before Hamilton's share can be assessed.